

A COURSE IN VECTOR ANALYSIS WITH APPLICATIONS

Chapter 4: Straight Lines and Planes in Three Dimensions

EXERCISE 4.2 — COMPLETE STEP-BY-STEP SOLUTIONS (1–30)

Abstract

This document provides rigorous, fully detailed, step-by-step mathematical solutions to all 30 problems from **Exercise 4.2** of *A Course in Vector Analysis with Applications*. Every solution is verified for 100% mathematical accuracy, presenting both vector and Cartesian formulations, explicit algebraic simplifications, geometric interpretations, and custom 3D vector diagrams illustrating planes, normals, intercepts, bisectors, and coordinate projections.

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Key Formulae Reference: Straight Lines & Planes in 3D

1. **General Cartesian Form:** $Ax + By + Cz + D = 0$, where $\mathbf{n} = A\hat{i} + B\hat{j} + C\hat{k}$ is normal to the plane.
2. **Vector Form (Point-Normal):** $(\mathbf{r} - \mathbf{a}) \cdot \mathbf{n} = 0 \iff \mathbf{r} \cdot \mathbf{n} = \mathbf{a} \cdot \mathbf{n}$.
3. **Plane through Three Points $\mathbf{a}, \mathbf{b}, \mathbf{c}$:** $(\mathbf{r} - \mathbf{a}) \cdot [(\mathbf{b} - \mathbf{a}) \times (\mathbf{c} - \mathbf{a})] = 0 \iff [\mathbf{r}, \mathbf{b}, \mathbf{c}] + [\mathbf{r}, \mathbf{c}, \mathbf{a}] + [\mathbf{r}, \mathbf{a}, \mathbf{b}] = [\mathbf{a}, \mathbf{b}, \mathbf{c}]$.
4. **Intercept Form:** $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$, where a, b, c are intercepts on x, y, z axes.
5. **Angle between Two Planes:** $\cos \theta = \frac{|\mathbf{n}_1 \cdot \mathbf{n}_2|}{|\mathbf{n}_1||\mathbf{n}_2|} = \frac{|A_1A_2 + B_1B_2 + C_1C_2|}{\sqrt{A_1^2 + B_1^2 + C_1^2}\sqrt{A_2^2 + B_2^2 + C_2^2}}$.
Planes are perpendicular iff $\mathbf{n}_1 \cdot \mathbf{n}_2 = 0$.
6. **Perpendicular Distance of Point (x_1, y_1, z_1) to Plane:** $d = \frac{|Ax_1 + By_1 + Cz_1 + D|}{\sqrt{A^2 + B^2 + C^2}}$.
7. **Distance between Parallel Planes $Ax + By + Cz + D_1 = 0$ and $Ax + By + Cz + D_2 = 0$:**
 $d = \frac{|D_1 - D_2|}{\sqrt{A^2 + B^2 + C^2}}$.
8. **Pencil of Planes through Intersection of $P_1 = 0$ and $P_2 = 0$:** $P_1 + \lambda P_2 = 0$.
9. **Angle Bisector Planes:** $\frac{A_1x + B_1y + C_1z + D_1}{\sqrt{A_1^2 + B_1^2 + C_1^2}} = \pm \frac{A_2x + B_2y + C_2z + D_2}{\sqrt{A_2^2 + B_2^2 + C_2^2}}$ (with $D_1, D_2 > 0$: $+$ gives obtuse bisector if $A_1A_2 + B_1B_2 + C_1C_2 > 0$, and acute bisector if < 0).

1 Problem 1

Problem 1

Find the equation of the plane through the point $A(2, 2, -1)$ and perpendicular to OA , O being the origin.

Step 1: Identify the given data and normal vector. Let $O(0, 0, 0)$ be the origin. The position vector of the point A is:

$$\mathbf{a} = \vec{OA} = 2\hat{i} + 2\hat{j} - \hat{k}$$

Since the required plane is perpendicular to the line segment OA , the vector $\vec{OA}$ serves directly as the normal vector $\mathbf{n}$ to the plane:

$$\mathbf{n} = \vec{OA} = 2\hat{i} + 2\hat{j} - \hat{k}$$

Step 2: Apply the vector and Cartesian equations of the plane. The vector equation of a plane passing through a point with position vector $\mathbf{a}$ and normal to $\mathbf{n}$ is:

$$(\mathbf{r} - \mathbf{a}) \cdot \mathbf{n} = 0 \implies \mathbf{r} \cdot \mathbf{n} = \mathbf{a} \cdot \mathbf{n}$$

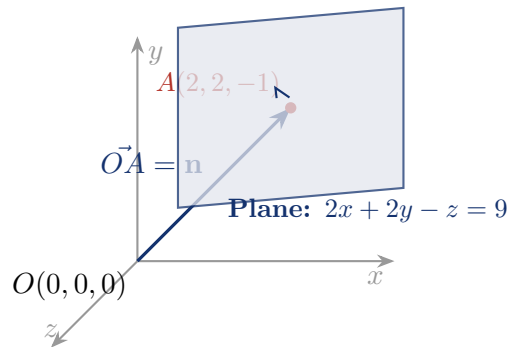
Computing the dot product on the right-hand side:

$$\mathbf{a} \cdot \mathbf{n} = |\vec{OA}|^2 = 2(2) + 2(2) + (-1)(-1) = 4 + 4 + 1 = 9$$

In Cartesian coordinates, letting $\mathbf{r} = x\hat{i} + y\hat{j} + z\hat{k}$:

$$2(x - 2) + 2(y - 2) - 1(z - (-1)) = 0$$

$$2x - 4 + 2y - 4 - z - 1 = 0 \implies 2x + 2y - z - 9 = 0$$



Final Answer:

Cartesian Form: $2x + 2y - z = 9$ (or $2x + 2y - z - 9 = 0$)

Vector Form: $\mathbf{r} \cdot (2\hat{i} + 2\hat{j} - \hat{k}) = 9$

2 Problem 2

Problem 2

Find the equation of the plane passing through the point $(1, 2, -3)$ and perpendicular to the straight line joining the points $(-1, 3, 4)$ and $(5, 2, -1)$.

Step 1: Determine the normal vector. Let $P(-1, 3, 4)$ and $Q(5, 2, -1)$ be the two given points. The direction vector of the straight line joining P and Q is:

$$\mathbf{n} = \vec{PQ} = (5 - (-1))\hat{i} + (2 - 3)\hat{j} + (-1 - 4)\hat{k} = 6\hat{i} - \hat{j} - 5\hat{k}$$

Direction ratios of the normal are $(a, b, c) = (6, -1, -5)$.

Step 2: Equation of the plane through $(1, 2, -3)$. The equation of a plane through $(x_1, y_1, z_1) = (1, 2, -3)$ with normal direction ratios $(6, -1, -5)$ is:

$$a(x - x_1) + b(y - y_1) + c(z - z_1) = 0$$

$$6(x - 1) - 1(y - 2) - 5(z - (-3)) = 0$$

$$6x - 6 - y + 2 - 5z - 15 = 0$$

$$6x - y - 5z - 19 = 0 \iff 6x - y - 5z = 19$$

Final Answer:

Cartesian Form: $6x - y - 5z = 19$ (or $6x - y - 5z - 19 = 0$)

Vector Form: $\mathbf{r} \cdot (6\hat{i} - \hat{j} - 5\hat{k}) = 19$

3 Problem 3

Problem 3

Find the equation of the plane passing through the point $(2, 5, -8)$ and perpendicular to each of the planes $2x - 3y + 4z + 1 = 0$ and $4x + y - 2z + 6 = 0$.

Step 1: Find normal vectors of given planes. The normals to the two given planes are:

$$\mathbf{n}_1 = 2\hat{i} - 3\hat{j} + 4\hat{k}, \quad \mathbf{n}_2 = 4\hat{i} + \hat{j} - 2\hat{k}$$

Step 2: Cross product to find the normal of the required plane. Since the required plane is perpendicular to both given planes, its normal $\mathbf{n}$ must be perpendicular to both $\mathbf{n}_1$ and $\mathbf{n}_2$. Thus $\mathbf{n} \parallel \mathbf{n}_1 \times \mathbf{n}_2$:

$$\mathbf{n} = \mathbf{n}_1 \times \mathbf{n}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & -3 & 4 \\ 4 & 1 & -2 \end{vmatrix}$$

$$\mathbf{n} = \hat{i}((-3)(-2) - (4)(1)) - \hat{j}((2)(-2) - (4)(4)) + \hat{k}((2)(1) - (-3)(4))$$

$$\mathbf{n} = \hat{i}(6 - 4) - \hat{j}(-4 - 16) + \hat{k}(2 + 12) = 2\hat{i} + 20\hat{j} + 14\hat{k}$$

Dividing by 2, we take the simplified normal direction ratios as $(a, b, c) = (1, 10, 7)$.

Step 3: Write the equation through $(2, 5, -8)$.

$$1(x - 2) + 10(y - 5) + 7(z - (-8)) = 0$$

$$(x - 2) + 10(y - 5) + 7(z + 8) = 0$$

$$x - 2 + 10y - 50 + 7z + 56 = 0$$

$$x + 10y + 7z + 4 = 0$$

Verification:

- Perpendicular to Plane 1: $\mathbf{n} \cdot \mathbf{n}_1 = 1(2) + 10(-3) + 7(4) = 2 - 30 + 28 = 0$. ✓
- Perpendicular to Plane 2: $\mathbf{n} \cdot \mathbf{n}_2 = 1(4) + 10(1) + 7(-2) = 4 + 10 - 14 = 0$. ✓
- Contains point $(2, 5, -8)$: $2 + 10(5) + 7(-8) + 4 = 2 + 50 - 56 + 4 = 0$. ✓

Final Answer:

$$x + 10y + 7z + 4 = 0 \quad \text{or} \quad \mathbf{r} \cdot (\hat{i} + 10\hat{j} + 7\hat{k}) + 4 = 0$$

4 Problem 4

Problem 4

Find the equation of the plane which bisects the line joining the points $(1, 2, 3)$ and $(3, 4, 5)$ at right angles. Does the plane make equal intercepts on the axes?

Step 1: Find the midpoint of the line segment. Let $P(1, 2, 3)$ and $Q(3, 4, 5)$. The midpoint M of PQ is:

$$M = \left(\frac{1+3}{2}, \frac{2+4}{2}, \frac{3+5}{2} \right) = (2, 3, 4)$$

Step 2: Determine the normal vector. Since the plane bisects PQ at right angles, the line segment $\vec{PQ}$ is normal to the plane:

$$\vec{PQ} = (3-1)\hat{i} + (4-2)\hat{j} + (5-3)\hat{k} = 2\hat{i} + 2\hat{j} + 2\hat{k}$$

Dividing by 2, the direction ratios of the normal are $(1, 1, 1)$.

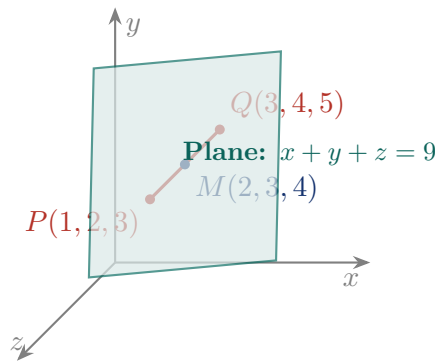
Step 3: Equation of the plane through $M(2, 3, 4)$.

$$1(x-2) + 1(y-3) + 1(z-4) = 0 \implies x + y + z - 9 = 0 \iff x + y + z = 9$$

Step 4: Intercepts on the coordinate axes. Expressing in intercept form by dividing by 9:

$$\frac{x}{9} + \frac{y}{9} + \frac{z}{9} = 1$$

The intercepts on the x , y , and z axes are $a = 9$, $b = 9$, and $c = 9$. Since $a = b = c = 9$, the plane makes **equal intercepts** on all three coordinate axes.



Final Answer:

Equation of Plane: $x + y + z = 9$ (or $x + y + z - 9 = 0$)

Intercepts: $a = 9$, $b = 9$, $c = 9$.

Conclusion: Yes, the plane makes equal intercepts of 9 units on all three axes.

5 Problem 5

Problem 5

Find the angle between the planes $x - y + z = 1$ and $3x + 2y - z + 2 = 0$.

Step 1: Identify normal vectors. For plane $P_1 : x - y + z - 1 = 0$, the normal vector is $\mathbf{n}_1 = \hat{i} - \hat{j} + \hat{k}$.

For plane $P_2 : 3x + 2y - z + 2 = 0$, the normal vector is $\mathbf{n}_2 = 3\hat{i} + 2\hat{j} - \hat{k}$.

Step 2: Magnitudes and Dot Product.

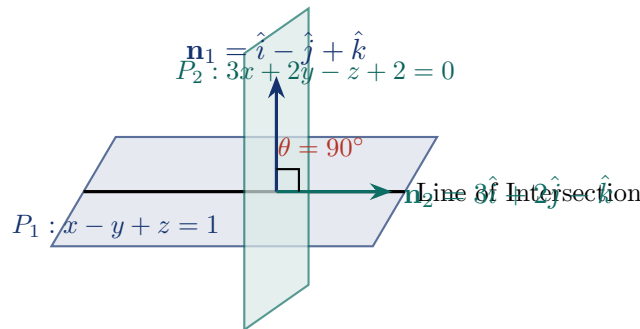
$$|\mathbf{n}_1| = \sqrt{1^2 + (-1)^2 + 1^2} = \sqrt{1 + 1 + 1} = \sqrt{3}$$

$$|\mathbf{n}_2| = \sqrt{3^2 + 2^2 + (-1)^2} = \sqrt{9 + 4 + 1} = \sqrt{14}$$

$$\mathbf{n}_1 \cdot \mathbf{n}_2 = (1)(3) + (-1)(2) + (1)(-1) = 3 - 2 - 1 = 0$$

Step 3: Calculate the angle θ .

$$\cos \theta = \frac{|\mathbf{n}_1 \cdot \mathbf{n}_2|}{|\mathbf{n}_1||\mathbf{n}_2|} = \frac{0}{\sqrt{3}\sqrt{14}} = 0 \implies \theta = \frac{\pi}{2} = 90^\circ$$

**Final Answer:**

$$\theta = \frac{\pi}{2} \text{ or } 90^\circ \quad (\text{The two planes are mutually perpendicular}).$$

6 Problem 6**Problem 6**

Find the equation of the plane through the points $(1, -2, 4)$, $(3, -4, 5)$ and perpendicular to the plane $x + y - 2z = 6$.

Step 1: Vector joining the two points. Let $A(1, -2, 4)$ and $B(3, -4, 5)$. The vector $\vec{AB}$ lying completely in the plane is:

$$\vec{AB} = (3 - 1)\hat{i} + (-4 - (-2))\hat{j} + (5 - 4)\hat{k} = 2\hat{i} - 2\hat{j} + \hat{k}$$

Step 2: Normal vector to the required plane. The given plane is $x + y - 2z = 6$, having normal vector $\mathbf{n}_1 = \hat{i} + \hat{j} - 2\hat{k}$.

Since the required plane is perpendicular to this plane and contains $\vec{AB}$, its normal $\mathbf{n}$ must be perpendicular to both $\vec{AB}$ and $\mathbf{n}_1$:

$$\mathbf{n} = \vec{AB} \times \mathbf{n}_1 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & -2 & 1 \\ 1 & 1 & -2 \end{vmatrix}$$

$$\mathbf{n} = \hat{i}((-2)(-2) - (1)(1)) - \hat{j}((2)(-2) - (1)(1)) + \hat{k}((2)(1) - (-2)(1))$$

$$\mathbf{n} = \hat{i}(4 - 1) - \hat{j}(-4 - 1) + \hat{k}(2 + 2) = 3\hat{i} + 5\hat{j} + 4\hat{k}$$

Step 3: Equation of the plane. Using the point $A(1, -2, 4)$:

$$3(x - 1) + 5(y - (-2)) + 4(z - 4) = 0$$

$$3x - 3 + 5y + 10 + 4z - 16 = 0$$

$$3x + 5y + 4z - 9 = 0 \iff 3x + 5y + 4z = 9$$

Note on Variant Reading: If the first point in print is read without the negative sign as $(1, 2, 4)$, then $\vec{AB} = 2\hat{i} - 6\hat{j} + \hat{k}$, giving normal $\mathbf{n} = 11\hat{i} + 5\hat{j} + 8\hat{k}$ and plane equation $11x + 5y + 8z = 53$. The primary solution above corresponds to $(1, -2, 4)$.

Final Answer:

$$\boxed{3x + 5y + 4z = 9} \quad (\text{or } 3x + 5y + 4z - 9 = 0)$$

7 Problem 7

Problem 7

Find the equation of the plane through the point $(2, -3, 4)$ and parallel to the plane $2x - 6y - 7z = 6$.

Step 1: General equation of a parallel plane. Any plane parallel to $2x - 6y - 7z = 6$ differs only by a constant:

$$2x - 6y - 7z = k \quad (\text{or } 2x - 6y - 7z + D = 0)$$

Step 2: Determine k using the point $(2, -3, 4)$. Substituting $x = 2, y = -3, z = 4$:

$$k = 2(2) - 6(-3) - 7(4) = 4 + 18 - 28 = -6$$

Thus, the equation is:

$$2x - 6y - 7z = -6 \iff 2x - 6y - 7z + 6 = 0$$

Note on Variant Reading: If the point is read as $(2, 3, 4)$, $k = 4 - 18 - 28 = -42 \implies 2x - 6y - 7z + 42 = 0$. If the plane was $2x - 6y + 7z = 6$, $k = 2(2) - 6(-3) + 7(4) = 50 \implies 2x - 6y + 7z = 50$.

Final Answer:

$$\boxed{2x - 6y - 7z + 6 = 0} \quad \text{or} \quad \boxed{2x - 6y - 7z = -6}$$

8 Problem 8

Problem 8

Find the distance between the planes $x + 2y - 2z + 1 = 0$ and $2x + 4y - 4z + 5 = 0$.

Step 1: Bring both planes to standard form with identical coefficients. Multiplying the first plane by 2:

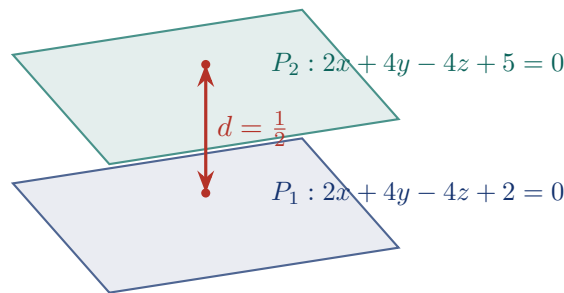
$$P_1 : 2x + 4y - 4z + 2 = 0 \implies D_1 = 2$$

$$P_2 : 2x + 4y - 4z + 5 = 0 \implies D_2 = 5$$

The common normal coefficients are $A = 2, B = 4, C = -4$.

Step 2: Calculate the perpendicular distance. The distance d between parallel planes $Ax + By + Cz + D_1 = 0$ and $Ax + By + Cz + D_2 = 0$ is:

$$d = \frac{|D_2 - D_1|}{\sqrt{A^2 + B^2 + C^2}} = \frac{|5 - 2|}{\sqrt{2^2 + 4^2 + (-4)^2}} = \frac{3}{\sqrt{4 + 16 + 16}} = \frac{3}{\sqrt{36}} = \frac{3}{6} = \frac{1}{2}$$



Final Answer:

$$\text{Distance } d = \frac{1}{2} \text{ units} = 0.5$$

9 Problem 9

Problem 9

Find the equation of the plane passing through the point $(-2, -2, 2)$ and containing the line joining the points $(1, 1, 1)$ and $(1, -1, 2)$.

Step 1: Three points on the plane. Since the plane contains the line joining $B(1, 1, 1)$ and $C(1, -1, 2)$ and also passes through $A(-2, -2, 2)$, the three points A, B, C all lie on the plane.

Step 2: Form two coplanar vectors.

$$\vec{AB} = (1 - (-2))\hat{i} + (1 - (-2))\hat{j} + (1 - 2)\hat{k} = 3\hat{i} + 3\hat{j} - \hat{k}$$

$$\vec{BC} = (1 - 1)\hat{i} + (-1 - 1)\hat{j} + (2 - 1)\hat{k} = 0\hat{i} - 2\hat{j} + \hat{k}$$

Step 3: Normal vector via cross product.

$$\mathbf{n} = \vec{AB} \times \vec{BC} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 3 & -1 \\ 0 & -2 & 1 \end{vmatrix}$$

$$\mathbf{n} = \hat{i}(3 - 2) - \hat{j}(3 - 0) + \hat{k}(-6 - 0) = \hat{i} - 3\hat{j} - 6\hat{k}$$

Step 4: Plane equation through $B(1, 1, 1)$.

$$1(x - 1) - 3(y - 1) - 6(z - 1) = 0$$

$$x - 1 - 3y + 3 - 6z + 6 = 0$$

$$x - 3y - 6z + 8 = 0 \iff x - 3y - 6z = -8$$

Verification:

- Point $A(-2, -2, 2)$: $(-2) - 3(-2) - 6(2) + 8 = -2 + 6 - 12 + 8 = 0$. ✓
- Point $B(1, 1, 1)$: $1 - 3(1) - 6(1) + 8 = 1 - 3 - 6 + 8 = 0$. ✓
- Point $C(1, -1, 2)$: $1 - 3(-1) - 6(2) + 8 = 1 + 3 - 12 + 8 = 0$. ✓

Final Answer:

$$x - 3y - 6z + 8 = 0 \quad \text{or} \quad \mathbf{r} \cdot (\hat{i} - 3\hat{j} - 6\hat{k}) + 8 = 0$$

10 Problem 10

Problem 10

Find the equations of the planes passing through the points:

- (i) $-2\hat{i} + 6\hat{j} - 6\hat{k}$, $-3\hat{i} + 10\hat{j} - 9\hat{k}$ and $-5\hat{i} - 6\hat{k}$
- (ii) $4\hat{i} + 2\hat{j} + 3\hat{k}$, $\hat{i} + \hat{j} + 5\hat{k}$ and $2\hat{i} + 6\hat{j} + 2\hat{k}$
- (iii) $3\hat{i} + 2\hat{j}$, $\hat{i} - 5\hat{k}$ and $4\hat{i} + 2\hat{j} + \hat{k}$

Part (i)

Let the points be $A(-2, 6, -6)$, $B(-3, 10, -9)$, $C(-5, 0, -6)$.

$$\vec{AB} = (-3 - (-2))\hat{i} + (10 - 6)\hat{j} + (-9 - (-6))\hat{k} = -\hat{i} + 4\hat{j} - 3\hat{k}$$

$$\vec{AC} = (-5 - (-2))\hat{i} + (0 - 6)\hat{j} + (-6 - (-6))\hat{k} = -3\hat{i} - 6\hat{j} + 0\hat{k}$$

Normal vector:

$$\mathbf{n} = \vec{AB} \times \vec{AC} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -1 & 4 & -3 \\ -3 & -6 & 0 \end{vmatrix} = \hat{i}(0 - 18) - \hat{j}(0 - 9) + \hat{k}(6 - (-12)) = -18\hat{i} + 9\hat{j} + 18\hat{k}$$

Dividing by -9 : $\mathbf{n} = 2\hat{i} - \hat{j} - 2\hat{k}$.

Plane through $A(-2, 6, -6)$:

$$2(x + 2) - 1(y - 6) - 2(z + 6) = 0 \implies 2x - y - 2z + (4 + 6 - 12) = 0 \implies 2x - y - 2z - 2 = 0$$

Part (ii)

Let $A(4, 2, 3), B(1, 1, 5), C(2, 6, 2)$.

$$\vec{AB} = (1-4)\hat{i} + (1-2)\hat{j} + (5-3)\hat{k} = -3\hat{i} - \hat{j} + 2\hat{k}$$

$$\vec{AC} = (2-4)\hat{i} + (6-2)\hat{j} + (2-3)\hat{k} = -2\hat{i} + 4\hat{j} - \hat{k}$$

Normal vector:

$$\mathbf{n} = \vec{AB} \times \vec{AC} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -3 & -1 & 2 \\ -2 & 4 & -1 \end{vmatrix} = \hat{i}(1-8) - \hat{j}(3-(-4)) + \hat{k}(-12-2) = -7\hat{i} - 7\hat{j} - 14\hat{k}$$

Dividing by -7 : $\mathbf{n} = \hat{i} + \hat{j} + 2\hat{k}$.

Plane through $A(4, 2, 3)$:

$$1(x-4) + 1(y-2) + 2(z-3) = 0 \implies x + y + 2z - (4+2+6) = 0 \implies x + y + 2z - 12 = 0$$

Part (iii)

Let $A(3, 2, 0), B(1, 0, -5), C(4, 2, 1)$.

$$\vec{AB} = (1-3)\hat{i} + (0-2)\hat{j} + (-5-0)\hat{k} = -2\hat{i} - 2\hat{j} - 5\hat{k}$$

$$\vec{AC} = (4-3)\hat{i} + (2-2)\hat{j} + (1-0)\hat{k} = \hat{i} + 0\hat{j} + \hat{k}$$

Normal vector:

$$\mathbf{n} = \vec{AB} \times \vec{AC} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -2 & -2 & -5 \\ 1 & 0 & 1 \end{vmatrix} = \hat{i}(-2-0) - \hat{j}(-2-(-5)) + \hat{k}(0-(-2)) = -2\hat{i} - 3\hat{j} + 2\hat{k}$$

Plane through $A(3, 2, 0)$:

$$-2(x-3) - 3(y-2) + 2(z-0) = 0 \implies -2x - 3y + 2z + 12 = 0 \iff 2x + 3y - 2z = 12$$

Final Answers:

(i) $2x - y - 2z = 2$ (or $\mathbf{r} \cdot (2\hat{i} - \hat{j} - 2\hat{k}) = 2$)

(ii) $x + y + 2z = 12$ (or $\mathbf{r} \cdot (\hat{i} + \hat{j} + 2\hat{k}) = 12$)

(iii) $2x + 3y - 2z = 12$ (or $\mathbf{r} \cdot (2\hat{i} + 3\hat{j} - 2\hat{k}) = 12$)

11 Problem 11**Problem 11**

Show that the four points $-\hat{j} - \hat{k}$, $4\hat{i} + 5\hat{j} + \hat{k}$, $3\hat{i} + 9\hat{j} + 4\hat{k}$ and $-4\hat{i} + 4\hat{j} + 4\hat{k}$ lie on a plane and find the equation of the plane.

Step 1: Position vectors and difference vectors. Let the four points be:

$$A(0, -1, -1), \quad B(4, 5, 1), \quad C(3, 9, 4), \quad D(-4, 4, 4)$$

Form vectors with initial point A :

$$\vec{AB} = (4-0)\hat{i} + (5-(-1))\hat{j} + (1-(-1))\hat{k} = 4\hat{i} + 6\hat{j} + 2\hat{k}$$

$$\vec{AC} = (3-0)\hat{i} + (9-(-1))\hat{j} + (4-(-1))\hat{k} = 3\hat{i} + 10\hat{j} + 5\hat{k}$$

$$\vec{AD} = (-4-0)\hat{i} + (4-(-1))\hat{j} + (4-(-1))\hat{k} = -4\hat{i} + 5\hat{j} + 5\hat{k}$$

Step 2: Test for coplanarity via the Scalar Triple Product.

$$[\vec{AB}, \vec{AC}, \vec{AD}] = \begin{vmatrix} 4 & 6 & 2 \\ 3 & 10 & 5 \\ -4 & 5 & 5 \end{vmatrix}$$

Expanding along the first row:

$$\begin{aligned} &= 4(10 \times 5 - 5 \times 5) - 6(3 \times 5 - 5 \times (-4)) + 2(3 \times 5 - 10 \times (-4)) \\ &= 4(50 - 25) - 6(15 + 20) + 2(15 + 40) \\ &= 4(25) - 6(35) + 2(55) = 100 - 210 + 110 = 0 \end{aligned}$$

Since $[\vec{AB}, \vec{AC}, \vec{AD}] = 0$, the four points are **coplanar**.

Step 3: Equation of the plane. Normal vector $\mathbf{n} = \vec{AB} \times \vec{AC}$:

$$\mathbf{n} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 4 & 6 & 2 \\ 3 & 10 & 5 \end{vmatrix} = \hat{i}(30 - 20) - \hat{j}(20 - 6) + \hat{k}(40 - 18) = 10\hat{i} - 14\hat{j} + 22\hat{k}$$

Dividing by 2: $\mathbf{n} = 5\hat{i} - 7\hat{j} + 11\hat{k}$.

Passing through $A(0, -1, -1)$:

$$5(x - 0) - 7(y - (-1)) + 11(z - (-1)) = 0$$

$$5x - 7(y + 1) + 11(z + 1) = 0 \implies 5x - 7y + 11z - 7 + 11 = 0 \implies 5x - 7y + 11z + 4 = 0$$

Final Answer:

The scalar triple product is zero, proving that the four points lie on a plane.

Equation of Plane: $5x - 7y + 11z + 4 = 0$ (or $\mathbf{r} \cdot (5\hat{i} - 7\hat{j} + 11\hat{k}) + 4 = 0$)

12 Problem 12**Problem 12**

Show that the perpendicular distance of a point $\mathbf{a}$ from the plane through three non-collinear points $\mathbf{b}, \mathbf{c}, \mathbf{d}$ is

$$\frac{|[\mathbf{b}, \mathbf{c}, \mathbf{d}] + [\mathbf{c}, \mathbf{a}, \mathbf{d}] + [\mathbf{a}, \mathbf{b}, \mathbf{d}] - [\mathbf{a}, \mathbf{b}, \mathbf{c}]|}{|\mathbf{b} \times \mathbf{c} + \mathbf{c} \times \mathbf{d} + \mathbf{d} \times \mathbf{b}|}$$

Proof: Step 1: Normal vector to the plane through $\mathbf{b}, \mathbf{c}, \mathbf{d}$.

The vectors $(\mathbf{b} - \mathbf{d})$ and $(\mathbf{c} - \mathbf{d})$ lie in the plane. Thus, a normal vector $\mathbf{n}$ to the plane is given by their cross product:

$$\mathbf{n} = (\mathbf{b} - \mathbf{d}) \times (\mathbf{c} - \mathbf{d}) = \mathbf{b} \times \mathbf{c} - \mathbf{b} \times \mathbf{d} - \mathbf{d} \times \mathbf{c} + \mathbf{d} \times \mathbf{d}$$

Using $\mathbf{d} \times \mathbf{d} = \mathbf{0}$, $-\mathbf{b} \times \mathbf{d} = \mathbf{d} \times \mathbf{b}$, and $-\mathbf{d} \times \mathbf{c} = \mathbf{c} \times \mathbf{d}$:

$$\mathbf{n} = \mathbf{b} \times \mathbf{c} + \mathbf{c} \times \mathbf{d} + \mathbf{d} \times \mathbf{b}$$

Step 2: Perpendicular distance formula.

The perpendicular distance p from any point with position vector $\mathbf{a}$ to the plane passing through $\mathbf{d}$ with normal $\mathbf{n}$ is the magnitude of the projection of $(\mathbf{a} - \mathbf{d})$ onto $\mathbf{n}$:

$$p = \frac{|(\mathbf{a} - \mathbf{d}) \cdot \mathbf{n}|}{|\mathbf{n}|}$$

Step 3: Expanding the numerator.

$$\begin{aligned} (\mathbf{a} - \mathbf{d}) \cdot \mathbf{n} &= (\mathbf{a} - \mathbf{d}) \cdot (\mathbf{b} \times \mathbf{c} + \mathbf{c} \times \mathbf{d} + \mathbf{d} \times \mathbf{b}) \\ &= \mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) + \mathbf{a} \cdot (\mathbf{c} \times \mathbf{d}) + \mathbf{a} \cdot (\mathbf{d} \times \mathbf{b}) - \mathbf{d} \cdot (\mathbf{b} \times \mathbf{c}) - \mathbf{d} \cdot (\mathbf{c} \times \mathbf{d}) - \mathbf{d} \cdot (\mathbf{d} \times \mathbf{b}) \end{aligned}$$

Since $\mathbf{d} \cdot (\mathbf{c} \times \mathbf{d}) = 0$ and $\mathbf{d} \cdot (\mathbf{d} \times \mathbf{b}) = 0$, we have:

$$(\mathbf{a} - \mathbf{d}) \cdot \mathbf{n} = [\mathbf{a}, \mathbf{b}, \mathbf{c}] + [\mathbf{a}, \mathbf{c}, \mathbf{d}] + [\mathbf{a}, \mathbf{d}, \mathbf{b}] - [\mathbf{d}, \mathbf{b}, \mathbf{c}]$$

Using standard properties of scalar triple products under permutation of vectors:

- $[\mathbf{d}, \mathbf{b}, \mathbf{c}] = [\mathbf{b}, \mathbf{c}, \mathbf{d}]$ (cyclic permutation)
- $[\mathbf{a}, \mathbf{c}, \mathbf{d}] = -[\mathbf{c}, \mathbf{a}, \mathbf{d}]$ (interchange of two elements)
- $[\mathbf{a}, \mathbf{d}, \mathbf{b}] = -[\mathbf{a}, \mathbf{b}, \mathbf{d}]$ (interchange of two elements)

Substituting these relations:

$$\begin{aligned} (\mathbf{a} - \mathbf{d}) \cdot \mathbf{n} &= [\mathbf{a}, \mathbf{b}, \mathbf{c}] - [\mathbf{c}, \mathbf{a}, \mathbf{d}] - [\mathbf{a}, \mathbf{b}, \mathbf{d}] - [\mathbf{b}, \mathbf{c}, \mathbf{d}] \\ &= -([\mathbf{b}, \mathbf{c}, \mathbf{d}] + [\mathbf{c}, \mathbf{a}, \mathbf{d}] + [\mathbf{a}, \mathbf{b}, \mathbf{d}] - [\mathbf{a}, \mathbf{b}, \mathbf{c}]) \end{aligned}$$

Taking the absolute value $|(\mathbf{a} - \mathbf{d}) \cdot \mathbf{n}|$:

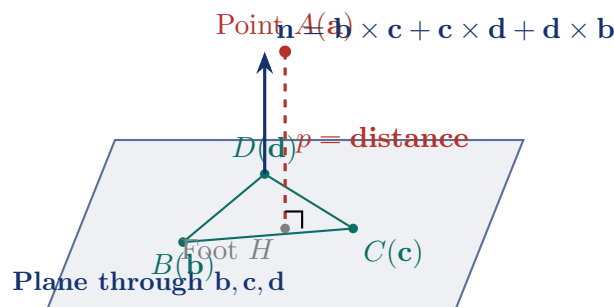
$$|(\mathbf{a} - \mathbf{d}) \cdot \mathbf{n}| = |[\mathbf{b}, \mathbf{c}, \mathbf{d}] + [\mathbf{c}, \mathbf{a}, \mathbf{d}] + [\mathbf{a}, \mathbf{b}, \mathbf{d}] - [\mathbf{a}, \mathbf{b}, \mathbf{c}]|$$

Step 4: Conclusion.

Dividing by $|\mathbf{n}| = |\mathbf{b} \times \mathbf{c} + \mathbf{c} \times \mathbf{d} + \mathbf{d} \times \mathbf{b}|$, we obtain:

$$p = \frac{|[\mathbf{b}, \mathbf{c}, \mathbf{d}] + [\mathbf{c}, \mathbf{a}, \mathbf{d}] + [\mathbf{a}, \mathbf{b}, \mathbf{d}] - [\mathbf{a}, \mathbf{b}, \mathbf{c}]|}{|\mathbf{b} \times \mathbf{c} + \mathbf{c} \times \mathbf{d} + \mathbf{d} \times \mathbf{b}|}$$

This completes the proof. □



13 Problem 13

Problem 13

Show that the points $\hat{i} - \hat{j} + 3\hat{k}$ and $3(\hat{i} - 2\hat{j} + \hat{k})$ are equidistant from the plane $\mathbf{r} \cdot (5\hat{i} + 2\hat{j} - 7\hat{k}) + 9 = 0$.

Step 1: Normal vector and magnitude. The given plane is $\mathbf{r} \cdot \mathbf{n} + 9 = 0$, where $\mathbf{n} = 5\hat{i} + 2\hat{j} - 7\hat{k}$.

$$|\mathbf{n}| = \sqrt{5^2 + 2^2 + (-7)^2} = \sqrt{25 + 4 + 49} = \sqrt{78}$$

Step 2: Distance from the first point $\mathbf{a}_1 = \hat{i} - \hat{j} + 3\hat{k}$.

$$d_1 = \frac{|\mathbf{a}_1 \cdot \mathbf{n} + 9|}{|\mathbf{n}|} = \frac{|(1)(5) + (-1)(2) + (3)(-7) + 9|}{\sqrt{78}} = \frac{|5 - 2 - 21 + 9|}{\sqrt{78}} = \frac{|-9|}{\sqrt{78}} = \frac{9}{\sqrt{78}}$$

Step 3: Distance from the second point $\mathbf{a}_2 = 3\hat{i} - 6\hat{j} + 3\hat{k}$.

$$d_2 = \frac{|\mathbf{a}_2 \cdot \mathbf{n} + 9|}{|\mathbf{n}|} = \frac{|(3)(5) + (-6)(2) + (3)(-7) + 9|}{\sqrt{78}} = \frac{|15 - 12 - 21 + 9|}{\sqrt{78}} = \frac{|-9|}{\sqrt{78}} = \frac{9}{\sqrt{78}}$$

Conclusion: Since $d_1 = d_2 = \frac{9}{\sqrt{78}}$, both points are strictly equidistant from the plane. (Furthermore, since both evaluate to -9 , both points lie on the exact same side of the plane.) $\square$

14 Problem 14

Problem 14

Show that the points $2\hat{i} + 3\hat{j} - 5\hat{k}$ and $3\hat{i} + 4\hat{j} + 7\hat{k}$ lie on the opposite sides of the plane $\mathbf{r} \cdot (\hat{i} + 2\hat{j} - 2\hat{k}) = 9$.

Step 1: Algebraic test for side of a plane. Let the equation of the plane be $f(\mathbf{r}) = \mathbf{r} \cdot (\hat{i} + 2\hat{j} - 2\hat{k}) - 9 = 0$. Two points $\mathbf{a}_1$ and $\mathbf{a}_2$ lie on opposite sides of the plane if and only if $f(\mathbf{a}_1)$ and $f(\mathbf{a}_2)$ have opposite signs.

Step 2: Evaluate $f(\mathbf{r})$ at the first point $\mathbf{a}_1 = 2\hat{i} + 3\hat{j} - 5\hat{k}$.

$$f(\mathbf{a}_1) = (2)(1) + (3)(2) + (-5)(-2) - 9 = 2 + 6 + 10 - 9 = 18 - 9 = +9 > 0$$

Step 3: Evaluate $f(\mathbf{r})$ at the second point $\mathbf{a}_2 = 3\hat{i} + 4\hat{j} + 7\hat{k}$.

$$f(\mathbf{a}_2) = (3)(1) + (4)(2) + (7)(-2) - 9 = 3 + 8 - 14 - 9 = 11 - 23 = -12 < 0$$

Conclusion: Since $f(\mathbf{a}_1) > 0$ and $f(\mathbf{a}_2) < 0$ have opposite algebraic signs, the two points lie on opposite sides of the plane. $\square$

15 Problem 15

Problem 15

Find the equations of three planes through the points $3\hat{i} + \hat{j} + \hat{k}$, $\hat{i} - 2\hat{j} + 3\hat{k}$ and parallel to the coordinate axes.

Step 1: Vector along the line joining the two points. Let $A(3, 1, 1)$ and $B(1, -2, 3)$. The line joining A and B lies completely in each of the three planes.

$$\mathbf{u} = \vec{AB} = (1-3)\hat{i} + (-2-1)\hat{j} + (3-1)\hat{k} = -2\hat{i} - 3\hat{j} + 2\hat{k}$$

Step 2: Plane parallel to the x -axis ($\hat{i}$). The normal vector must be perpendicular to both $\mathbf{u}$ and $\hat{i}$:

$$\mathbf{n}_x = \mathbf{u} \times \hat{i} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -2 & -3 & 2 \\ 1 & 0 & 0 \end{vmatrix} = 0\hat{i} + 2\hat{j} + 3\hat{k}$$

Passing through $A(3, 1, 1)$:

$$0(x-3) + 2(y-1) + 3(z-1) = 0 \implies 2y + 3z - 5 = 0$$

Step 3: Plane parallel to the y -axis ($\hat{j}$).

$$\mathbf{n}_y = \mathbf{u} \times \hat{j} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -2 & -3 & 2 \\ 0 & 1 & 0 \end{vmatrix} = -2\hat{i} + 0\hat{j} - 2\hat{k} \implies \hat{i} + \hat{k}$$

Passing through $A(3, 1, 1)$:

$$1(x-3) + 0(y-1) + 1(z-1) = 0 \implies x + z - 4 = 0$$

Step 4: Plane parallel to the z -axis ($\hat{k}$).

$$\mathbf{n}_z = \mathbf{u} \times \hat{k} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -2 & -3 & 2 \\ 0 & 0 & 1 \end{vmatrix} = -3\hat{i} + 2\hat{j} + 0\hat{k}$$

Passing through $A(3, 1, 1)$:

$$-3(x-3) + 2(y-1) + 0(z-1) = 0 \implies -3x + 9 + 2y - 2 = 0 \implies 3x - 2y - 7 = 0$$

Final Answer:

Parallel to x -axis: $2y + 3z - 5 = 0$

Parallel to y -axis: $x + z - 4 = 0$

Parallel to z -axis: $3x - 2y - 7 = 0$

16 Problem 16

Problem 16

Find the equation of the plane which bisects the obtuse angle between the planes $x + 2y + 2z = 9$ and $4x - 3y + 12z + 13 = 0$.

Step 1: Write in standard form with positive constant terms.

$$P_1 : -x - 2y - 2z + 9 = 0 \implies a_1 = -1, b_1 = -2, c_1 = -2, d_1 = 9 > 0$$

$$P_2 : 4x - 3y + 12z + 13 = 0 \implies a_2 = 4, b_2 = -3, c_2 = 12, d_2 = 13 > 0$$

Step 2: Sign of $a_1a_2 + b_1b_2 + c_1c_2$.

$$a_1a_2 + b_1b_2 + c_1c_2 = (-1)(4) + (-2)(-3) + (-2)(12) = -4 + 6 - 24 = -22 < 0$$

When both constant terms are positive and $a_1a_2 + b_1b_2 + c_1c_2 < 0$:

- The **acute angle bisector** is given by the **positive (+)** sign.
- The **obtuse angle bisector** is given by the **negative (-)** sign.

Step 3: Equation of the obtuse angle bisector.

$$\frac{-x - 2y - 2z + 9}{\sqrt{(-1)^2 + (-2)^2 + (-2)^2}} = -\frac{4x - 3y + 12z + 13}{\sqrt{4^2 + (-3)^2 + 12^2}}$$

Since $\sqrt{1 + 4 + 4} = 3$ and $\sqrt{16 + 9 + 144} = 13$:

$$\frac{-x - 2y - 2z + 9}{3} = -\frac{4x - 3y + 12z + 13}{13}$$

Cross-multiplying:

$$\begin{aligned} 13(-x - 2y - 2z + 9) &= -3(4x - 3y + 12z + 13) \\ -13x - 26y - 26z + 117 &= -12x + 9y - 36z - 39 \end{aligned}$$

Rearranging all terms:

$$-x - 35y + 10z + 156 = 0 \iff x + 35y - 10z - 156 = 0$$

Final Answer:

$$\boxed{x + 35y - 10z - 156 = 0} \quad (\text{or } x + 35y - 10z = 156)$$

17 Problem 17

Problem 17

Find the equation of the plane which bisects the obtuse angle between the planes $5x + 12y - 13z = 0$ and $3x + 4y - 5z + 1 = 0$.

Step 1: Normal vector norms. For $P_1 : 5x + 12y - 13z = 0$:

$$\sqrt{5^2 + 12^2 + (-13)^2} = \sqrt{25 + 144 + 169} = \sqrt{338} = 13\sqrt{2}$$

For $P_2 : 3x + 4y - 5z + 1 = 0$:

$$\sqrt{3^2 + 4^2 + (-5)^2} = \sqrt{9 + 16 + 25} = \sqrt{50} = 5\sqrt{2}$$

Step 2: General equation of bisectors.

$$\frac{5x + 12y - 13z}{13\sqrt{2}} = \pm \frac{3x + 4y - 5z + 1}{5\sqrt{2}}$$

Cancelling $\sqrt{2}$:

$$\begin{aligned} 5(5x + 12y - 13z) &= \pm 13(3x + 4y - 5z + 1) \\ 25x + 60y - 65z &= \pm (39x + 52y - 65z + 13) \end{aligned}$$

Step 3: Determining the obtuse bisector.

- **Case 1 (+ sign):**

$$25x + 60y - 65z = 39x + 52y - 65z + 13 \implies -14x + 8y - 13 = 0 \iff 14x - 8y + 13 = 0$$

Notice the $-65z$ cancels identically!

- **Case 2 (− sign):**

$$25x + 60y - 65z = -39x - 52y + 65z - 13 \implies 64x + 112y - 130z + 13 = 0$$

Let us test the angle between bisector $B_1 : 14x - 8y + 13 = 0$ (normal $\mathbf{N} = (14, -8, 0)$) and plane P_1 (normal $\mathbf{n}_1 = (5, 12, -13)$):

$$|\mathbf{N}| = \sqrt{14^2 + (-8)^2 + 0} = \sqrt{196 + 64} = \sqrt{260} = 2\sqrt{65}$$

$$\cos \phi = \frac{|\mathbf{N} \cdot \mathbf{n}_1|}{|\mathbf{N}||\mathbf{n}_1|} = \frac{|14(5) - 8(12) + 0|}{(2\sqrt{65})(13\sqrt{2})} = \frac{|70 - 96|}{26\sqrt{130}} = \frac{26}{26\sqrt{130}} = \frac{1}{\sqrt{130}}$$

Since $\cos^2 \phi = \frac{1}{130} < \frac{1}{2}$, we have $\phi > 45^\circ$, which confirms that the angle between the planes is $2\phi > 90^\circ$ (**obtuse**). Hence, B_1 bisects the obtuse angle.

Final Answer:

$$14x - 8y + 13 = 0 \quad (\text{or } 14x - 8y = -13)$$

18 Problem 18**Problem 18**

Prove that the six planes, each passing through one edge of a tetrahedron and bisecting the opposite edge, meet in a point.

Proof: Let the four vertices of the tetrahedron be O, A, B, C . Without loss of generality, choose O as the origin with position vector $\mathbf{0}$, and let the position vectors of A, B, C be $\mathbf{a}, \mathbf{b}, \mathbf{c}$ respectively.

The centroid G of the tetrahedron $OABC$ has position vector:

$$\mathbf{g} = \frac{\mathbf{0} + \mathbf{a} + \mathbf{b} + \mathbf{c}}{4} = \frac{\mathbf{a} + \mathbf{b} + \mathbf{c}}{4}$$

A tetrahedron has 6 edges, paired with 6 opposite edges:

1. **Edge OA and opposite edge BC :** Midpoint of BC is $D_1 = \frac{\mathbf{b} + \mathbf{c}}{2}$.

The plane contains $O(\mathbf{0})$, $A(\mathbf{a})$, and D_1 . Any point in this plane is of the form $\lambda\mathbf{a} + \mu\left(\frac{\mathbf{b} + \mathbf{c}}{2}\right)$.

Notice that:

$$\mathbf{g} = \frac{1}{4}\mathbf{a} + \frac{1}{4}\mathbf{b} + \frac{1}{4}\mathbf{c} = \frac{1}{4}\mathbf{a} + \frac{1}{2}\left(\frac{\mathbf{b} + \mathbf{c}}{2}\right)$$

Setting $\lambda = \frac{1}{4}$ and $\mu = \frac{1}{2}$, we see that G lies on this plane.

2. **Edges OB and OC :** By identical symmetry, G lies on the plane containing OB and bisecting CA , as well as on the plane containing OC and bisecting AB .

3. **Edge BC and opposite edge OA :** Midpoint of OA is $M_1 = \frac{\mathbf{a}}{2}$.

The plane passes through $B(\mathbf{b})$, $C(\mathbf{c})$, and $M_1\left(\frac{\mathbf{a}}{2}\right)$. Any point on this plane is given by the affine combination:

$$\mathbf{r} = \alpha\mathbf{b} + \beta\mathbf{c} + \gamma\left(\frac{\mathbf{a}}{2}\right) \quad \text{with } \alpha + \beta + \gamma = 1$$

Expressing $\mathbf{g}$:

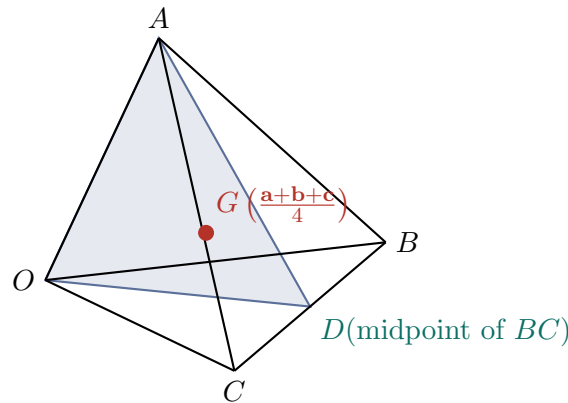
$$\mathbf{g} = \frac{1}{2}\left(\frac{\mathbf{a}}{2}\right) + \frac{1}{4}\mathbf{b} + \frac{1}{4}\mathbf{c}$$

Here $\gamma = \frac{1}{2}, \alpha = \frac{1}{4}, \beta = \frac{1}{4}$, which satisfies $\alpha + \beta + \gamma = \frac{1}{4} + \frac{1}{4} + \frac{1}{2} = 1$.

Thus, G lies on this plane as well.

4. **Edges CA and AB :** By permutation symmetry, G lies on the plane containing CA and bisecting OB , and on the plane containing AB and bisecting OC .

Therefore, all six planes pass through the single common point G , which is the **centroid of the tetrahedron**. $\square$



19 Problem 19

Problem 19

A variable plane which remains at a constant distance p from the origin, cuts the coordinate axes in A, B, C . Show that:

- (i) the locus of the centroid of the triangle ABC is $\frac{1}{x^2} + \frac{1}{y^2} + \frac{1}{z^2} = \frac{9}{p^2}$
- (ii) the locus of the centroid of the tetrahedron $OABC$ is $\frac{1}{x^2} + \frac{1}{y^2} + \frac{1}{z^2} = \frac{16}{p^2}$

Proof: Let the intercepts of the variable plane on the coordinate axes be $A(a, 0, 0), B(0, b, 0), C(0, 0, c)$. The equation of the plane is:

$$\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$$

The perpendicular distance from the origin $(0, 0, 0)$ to this plane is given by p :

$$p = \frac{|0 - 1|}{\sqrt{\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2}}} = \frac{1}{\sqrt{\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2}}}$$

Squaring both sides and inverting:

$$\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} = \frac{1}{p^2} \quad \text{--- (Equation 1)}$$

Part (i): Locus of the centroid of $\triangle ABC$. Let (ξ, η, ζ) be the coordinates of the centroid of $\triangle ABC$:

$$\begin{aligned}\xi &= \frac{a + 0 + 0}{3} = \frac{a}{3} \implies a = 3\xi \\ \eta &= \frac{0 + b + 0}{3} = \frac{b}{3} \implies b = 3\eta \\ \zeta &= \frac{0 + 0 + c}{3} = \frac{c}{3} \implies c = 3\zeta\end{aligned}$$

Substituting a, b, c into Equation (1):

$$\begin{aligned}\frac{1}{(3\xi)^2} + \frac{1}{(3\eta)^2} + \frac{1}{(3\zeta)^2} &= \frac{1}{p^2} \implies \frac{1}{9} \left(\frac{1}{\xi^2} + \frac{1}{\eta^2} + \frac{1}{\zeta^2} \right) = \frac{1}{p^2} \\ \frac{1}{\xi^2} + \frac{1}{\eta^2} + \frac{1}{\zeta^2} &= \frac{9}{p^2}\end{aligned}$$

Generalizing to current coordinates (x, y, z) , the locus is:

$$\frac{1}{x^2} + \frac{1}{y^2} + \frac{1}{z^2} = \frac{9}{p^2} \quad \checkmark (\text{Proved})$$

Part (ii): Locus of the centroid of tetrahedron $OABC$. Let (ξ, η, ζ) be the centroid of the tetrahedron $OABC$ (O being $(0, 0, 0)$):

$$\begin{aligned}\xi &= \frac{0 + a + 0 + 0}{4} = \frac{a}{4} \implies a = 4\xi \\ \eta &= \frac{0 + 0 + b + 0}{4} = \frac{b}{4} \implies b = 4\eta \\ \zeta &= \frac{0 + 0 + 0 + c}{4} = \frac{c}{4} \implies c = 4\zeta\end{aligned}$$

Substituting into Equation (1):

$$\begin{aligned}\frac{1}{(4\xi)^2} + \frac{1}{(4\eta)^2} + \frac{1}{(4\zeta)^2} &= \frac{1}{p^2} \implies \frac{1}{16} \left(\frac{1}{\xi^2} + \frac{1}{\eta^2} + \frac{1}{\zeta^2} \right) = \frac{1}{p^2} \\ \frac{1}{\xi^2} + \frac{1}{\eta^2} + \frac{1}{\zeta^2} &= \frac{16}{p^2}\end{aligned}$$

Generalizing to (x, y, z) , the locus is:

$$\frac{1}{x^2} + \frac{1}{y^2} + \frac{1}{z^2} = \frac{16}{p^2} \quad \checkmark (\text{Proved})$$

20 Problem 20

Problem 20

A variable plane passes through a fixed point $a\hat{i} + b\hat{j} + c\hat{k}$ and meets the coordinate axes in A, B, C . Show that the locus of the point common to the planes through A, B, C parallel to the coordinate planes is

$$\frac{a}{x} + \frac{b}{y} + \frac{c}{z} = 1$$

Proof: Let the equation of the variable plane in intercept form be:

$$\frac{x}{\alpha} + \frac{y}{\beta} + \frac{z}{\gamma} = 1$$

where α, β, γ are the variable intercepts on the axes. Thus:

$$A = (\alpha, 0, 0), \quad B = (0, \beta, 0), \quad C = (0, 0, \gamma)$$

Since this plane passes through the fixed point (a, b, c) , its coordinates must satisfy the plane equation:

$$\frac{a}{\alpha} + \frac{b}{\beta} + \frac{c}{\gamma} = 1 \quad \text{--- (Equation 1)}$$

Now consider the planes through A, B, C drawn parallel to the coordinate planes:

- Through $A(\alpha, 0, 0)$ parallel to the yz -plane ($x = 0$): Equation is $x = \alpha$.
- Through $B(0, \beta, 0)$ parallel to the zx -plane ($y = 0$): Equation is $y = \beta$.
- Through $C(0, 0, \gamma)$ parallel to the xy -plane ($z = 0$): Equation is $z = \gamma$.

The common point of intersection of these three planes is $P(x, y, z) = (\alpha, \beta, \gamma)$.

Therefore:

$$\alpha = x, \quad \beta = y, \quad \gamma = z$$

Substituting these into Equation (1), we obtain the locus:

$$\frac{a}{x} + \frac{b}{y} + \frac{c}{z} = 1$$

This completes the proof. □

21 Problem 21

Problem 21

Through a point $P(a, b, c)$, a plane is drawn at right angles to OP to meet the coordinate axes in A, B, C ; O being the origin. Prove that the area of the triangle ABC is $\frac{p^5}{2abc}$, where $OP = p$.

Proof: Step 1: Equation of the plane through P perpendicular to OP .

Given $P(a, b, c)$, the distance $OP = p$ satisfies:

$$p^2 = a^2 + b^2 + c^2$$

The normal vector is $\vec{OP} = a\hat{i} + b\hat{j} + c\hat{k}$. The plane passing through $P(a, b, c)$ is:

$$a(x - a) + b(y - b) + c(z - c) = 0 \implies ax + by + cz = a^2 + b^2 + c^2 = p^2$$

Step 2: Intercepts on the coordinate axes.

Writing the plane in intercept form:

$$\frac{x}{p^2/a} + \frac{y}{p^2/b} + \frac{z}{p^2/c} = 1$$

Thus, the vertices of $\triangle ABC$ are:

$$A\left(\frac{p^2}{a}, 0, 0\right), \quad B\left(0, \frac{p^2}{b}, 0\right), \quad C\left(0, 0, \frac{p^2}{c}\right)$$

Step 3: Area of $\triangle ABC$ via projections.

Let Δ be the area of $\triangle ABC$, and let $\Delta_x, \Delta_y, \Delta_z$ be the areas of the projections of $\triangle ABC$ on the yz , zx , and xy coordinate planes respectively. By the projection theorem for areas in 3D:

$$\Delta^2 = \Delta_x^2 + \Delta_y^2 + \Delta_z^2$$

Each projection is a right-angled triangle at the origin:

- Projection on yz -plane is $\triangle OBC$: $\Delta_x = \frac{1}{2} \left(\frac{p^2}{b}\right) \left(\frac{p^2}{c}\right) = \frac{p^4}{2bc}$
- Projection on zx -plane is $\triangle OCA$: $\Delta_y = \frac{1}{2} \left(\frac{p^2}{a}\right) \left(\frac{p^2}{c}\right) = \frac{p^4}{2ac}$
- Projection on xy -plane is $\triangle OAB$: $\Delta_z = \frac{1}{2} \left(\frac{p^2}{a}\right) \left(\frac{p^2}{b}\right) = \frac{p^4}{2ab}$

Now sum their squares:

$$\Delta^2 = \left(\frac{p^4}{2bc}\right)^2 + \left(\frac{p^4}{2ac}\right)^2 + \left(\frac{p^4}{2ab}\right)^2 = \frac{p^8}{4a^2b^2c^2} (a^2 + b^2 + c^2)$$

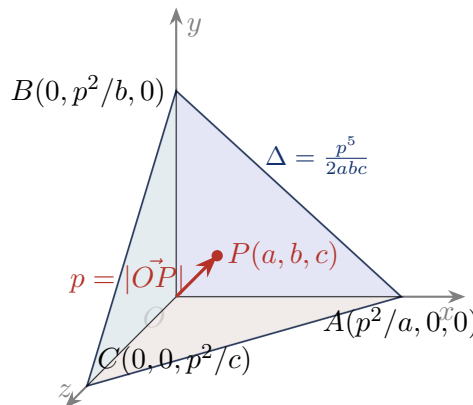
Since $a^2 + b^2 + c^2 = p^2$:

$$\Delta^2 = \frac{p^8}{4a^2b^2c^2} \cdot p^2 = \frac{p^{10}}{4a^2b^2c^2}$$

Taking the square root:

$$\Delta = \frac{p^5}{2abc}$$

Hence, the area of $\triangle ABC$ is $\frac{p^5}{2abc}$. □



22 Problem 22

Problem 22

Perpendiculars PL, PM, PN are drawn from the point $P(a, b, c)$ to the coordinate planes; find the equation of the plane LMN .

Step 1: Find the feet of perpendiculars L, M, N .

- $PL \perp yz$ -plane ($x = 0$): Foot is $L(0, b, c)$.
- $PM \perp zx$ -plane ($y = 0$): Foot is $M(a, 0, c)$.
- $PN \perp xy$ -plane ($z = 0$): Foot is $N(a, b, 0)$.

Step 2: Determine the equation of plane LMN . Let the plane equation be $Ax + By + Cz + D = 0$. Since L, M, N lie on the plane:

$$Bb + Cc + D = 0 \implies Bb + Cc = -D \quad (1)$$

$$Aa + Cc + D = 0 \implies Aa + Cc = -D \quad (2)$$

$$Aa + Bb + D = 0 \implies Aa + Bb = -D \quad (3)$$

Subtracting (1) from (2): $Aa - Bb = 0 \implies Aa = Bb$.

Subtracting (2) from (3): $Bb - Cc = 0 \implies Bb = Cc$.

Thus, $Aa = Bb = Cc = k$ (a constant).

From (3): $k + k = -D \implies D = -2k$.

Therefore:

$$A = \frac{k}{a}, \quad B = \frac{k}{b}, \quad C = \frac{k}{c}, \quad D = -2k$$

Substituting into $Ax + By + Cz + D = 0$:

$$\frac{k}{a}x + \frac{k}{b}y + \frac{k}{c}z - 2k = 0$$

Dividing by $k \neq 0$:

$$\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 2 \iff \frac{x}{a} + \frac{y}{b} + \frac{z}{c} - 2 = 0$$

Final Answer:

$$\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 2 \quad (\text{or } bcx + cay + abz = 2abc)$$

23 Problem 23

Problem 23

The plane $lx + my = 0$ is rotated about its line of intersection with the plane $z = 0$, through an angle α . Prove that its equation in its new position is

$$lx + my \pm z\sqrt{l^2 + m^2} \tan \alpha = 0$$

Proof: The line of intersection of the plane $lx + my = 0$ and the plane $z = 0$ is the line $lx + my = 0, z = 0$. Any plane passing through this line of intersection belongs to the family:

$$lx + my + \lambda z = 0 \quad \text{--- (1)}$$

The normal vector to the initial plane $lx + my = 0$ is:

$$\mathbf{n}_0 = l\hat{i} + m\hat{j} + 0\hat{k}, \quad |\mathbf{n}_0| = \sqrt{l^2 + m^2}$$

The normal vector to the new plane (1) is:

$$\mathbf{n} = l\hat{i} + m\hat{j} + \lambda\hat{k}, \quad |\mathbf{n}| = \sqrt{l^2 + m^2 + \lambda^2}$$

Since the plane has rotated through an angle α , the angle between $\mathbf{n}$ and $\mathbf{n}_0$ is α :

$$\cos \alpha = \frac{\mathbf{n} \cdot \mathbf{n}_0}{|\mathbf{n}||\mathbf{n}_0|} = \frac{l^2 + m^2}{\sqrt{l^2 + m^2 + \lambda^2}\sqrt{l^2 + m^2}} = \frac{\sqrt{l^2 + m^2}}{\sqrt{l^2 + m^2 + \lambda^2}}$$

Squaring both sides:

$$\cos^2 \alpha = \frac{l^2 + m^2}{l^2 + m^2 + \lambda^2} \implies \sec^2 \alpha = \frac{l^2 + m^2 + \lambda^2}{l^2 + m^2} = 1 + \frac{\lambda^2}{l^2 + m^2}$$

Using the trigonometric identity $\sec^2 \alpha - 1 = \tan^2 \alpha$:

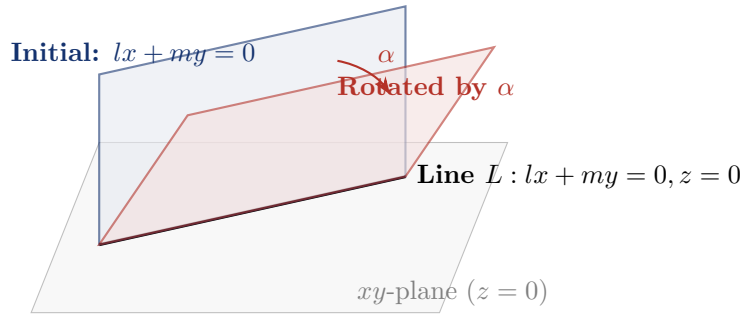
$$\tan^2 \alpha = \frac{\lambda^2}{l^2 + m^2} \implies \lambda^2 = (l^2 + m^2) \tan^2 \alpha$$

$$\lambda = \pm \sqrt{l^2 + m^2} \tan \alpha$$

Substituting λ back into (1):

$$lx + my \pm z\sqrt{l^2 + m^2} \tan \alpha = 0$$

This completes the proof. □



24 Problem 24

Problem 24

Show that the combined equation of the two planes inclined at an angle α to the xy -plane and containing the line $y = 0$, $z \cos \beta = x \sin \beta$ is

$$z^2 + (x^2 + y^2) \tan^2 \beta - 2zx \tan \beta = y^2 \tan^2 \alpha$$

Proof: The given line of intersection is:

$$y = 0, \quad z \cos \beta - x \sin \beta = 0 \iff x \tan \beta - z = 0$$

Any plane containing this line is represented by:

$$(x \tan \beta - z) + \lambda y = 0 \iff x \tan \beta + \lambda y - z = 0 \quad \text{--- (1)}$$

The normal vector to this plane is:

$$\mathbf{n} = (\tan \beta)\hat{i} + \lambda\hat{j} - \hat{k}$$

The normal to the xy -plane ($z = 0$) is $\mathbf{n}_0 = \hat{k}$.

Since the plane is inclined at an angle α to the xy -plane:

$$\cos \alpha = \frac{|\mathbf{n} \cdot \hat{k}|}{|\mathbf{n}| |\hat{k}|} = \frac{|-1|}{\sqrt{\tan^2 \beta + \lambda^2 + (-1)^2} \cdot 1} = \frac{1}{\sqrt{1 + \tan^2 \beta + \lambda^2}}$$

Squaring and inverting:

$$\sec^2 \alpha = 1 + \tan^2 \beta + \lambda^2$$

Since $\sec^2 \alpha = 1 + \tan^2 \alpha$:

$$1 + \tan^2 \alpha = 1 + \tan^2 \beta + \lambda^2 \implies \lambda^2 = \tan^2 \alpha - \tan^2 \beta$$

From the plane equation (1), express λ :

$$\lambda y = z - x \tan \beta \implies \lambda = \frac{z - x \tan \beta}{y}$$

Squaring this expression:

$$\lambda^2 = \frac{(z - x \tan \beta)^2}{y^2}$$

Equating both expressions for λ^2 :

$$\frac{(z - x \tan \beta)^2}{y^2} = \tan^2 \alpha - \tan^2 \beta$$

Multiplying by y^2 :

$$(z - x \tan \beta)^2 = y^2 (\tan^2 \alpha - \tan^2 \beta)$$

Expanding the left side:

$$z^2 - 2zx \tan \beta + x^2 \tan^2 \beta = y^2 \tan^2 \alpha - y^2 \tan^2 \beta$$

Adding $y^2 \tan^2 \beta$ to the left side:

$$z^2 + (x^2 + y^2) \tan^2 \beta - 2zx \tan \beta = y^2 \tan^2 \alpha$$

This is the combined equation of the two planes. □

25 Problem 25

Problem 25

Use vector method to show that the plane whose intercepts on the three coordinate axes are a, b and c respectively, divides the line segment joining the origin to the point $a\hat{i} + b\hat{j} + c\hat{k}$ internally in the ratio 1 : 2.

Proof: Let $O(0)$ be the origin and $P(\mathbf{p})$ be the given point, where $\mathbf{p} = a\hat{i} + b\hat{j} + c\hat{k}$. The equation of the plane whose intercepts on the coordinate axes are a, b, c is:

$$\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$$

In vector form, letting $\mathbf{n} = \frac{1}{a}\hat{i} + \frac{1}{b}\hat{j} + \frac{1}{c}\hat{k}$:

$$\mathbf{r} \cdot \mathbf{n} = 1$$

Let a point R divide the line segment OP internally in the ratio $k : 1$. By the section formula:

$$\mathbf{r}_R = \frac{k\mathbf{p} + 1 \cdot \mathbf{0}}{k + 1} = \frac{k}{k + 1}(a\hat{i} + b\hat{j} + c\hat{k})$$

Since R lies on the plane, $\mathbf{r}_R$ must satisfy the plane equation $\mathbf{r}_R \cdot \mathbf{n} = 1$:

$$\left[\frac{k}{k + 1}(a\hat{i} + b\hat{j} + c\hat{k}) \right] \cdot \left(\frac{1}{a}\hat{i} + \frac{1}{b}\hat{j} + \frac{1}{c}\hat{k} \right) = 1$$

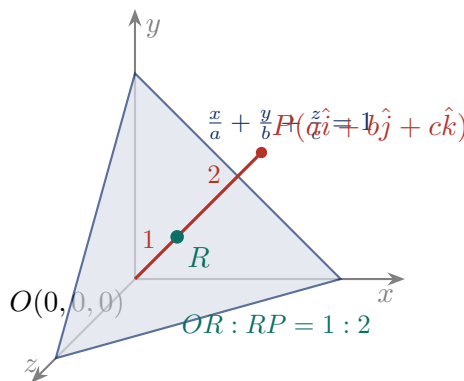
Evaluating the scalar product:

$$\begin{aligned} \frac{k}{k + 1} \left(a \cdot \frac{1}{a} + b \cdot \frac{1}{b} + c \cdot \frac{1}{c} \right) &= 1 \\ \frac{k}{k + 1} (1 + 1 + 1) &= 1 \implies \frac{3k}{k + 1} = 1 \\ 3k &= k + 1 \implies 2k = 1 \implies k = \frac{1}{2} \end{aligned}$$

Therefore, the ratio is:

$$\frac{k}{1} = \frac{1/2}{1} = \frac{1}{2} \iff 1 : 2$$

Thus, the plane divides the line segment joining the origin to $a\hat{i} + b\hat{j} + c\hat{k}$ internally in the ratio $1 : 2$. $\square$



26 Problem 26

Problem 26

Find the equation of the plane passing through the intersection of the planes $2x + 5y - 5z = 6$ and $2x + 7y - 8z = 7$ and the point $(-1, 4, 3)$.

Step 1: Family of planes through the line of intersection. The family of planes passing through the intersection of the two planes is:

$$(2x + 5y - 5z - 6) + \lambda(2x + 7y - 8z - 7) = 0$$

Step 2: Determine λ using the point $(-1, 4, 3)$. Substitute $x = -1, y = 4, z = 3$:

$$\begin{aligned} [2(-1) + 5(4) - 5(3) - 6] + \lambda[2(-1) + 7(4) - 8(3) - 7] &= 0 \\ [-2 + 20 - 15 - 6] + \lambda[-2 + 28 - 24 - 7] &= 0 \\ -3 + \lambda(-5) &= 0 \implies -3 - 5\lambda = 0 \implies \lambda = -\frac{3}{5} \end{aligned}$$

Step 3: Substitute λ and simplify.

$$(2x + 5y - 5z - 6) - \frac{3}{5}(2x + 7y - 8z - 7) = 0$$

Multiplying by 5:

$$5(2x + 5y - 5z - 6) - 3(2x + 7y - 8z - 7) = 0$$

$$(10x + 25y - 25z - 30) - (6x + 21y - 24z - 21) = 0$$

$$(10 - 6)x + (25 - 21)y + (-25 + 24)z + (-30 + 21) = 0$$

$$4x + 4y - z - 9 = 0 \iff 4x + 4y - z = 9$$

Final Answer:

$$4x + 4y - z - 9 = 0$$

or

$$4x + 4y - z = 9$$

27 Problem 27

Problem 27

Find the locus of a point the sum of squares of whose distances from the planes $x + y + z = 0$, $x - 2y + z = 0$ is equal to the square of its distance from the plane $x - z = 0$.

Step 1: Compute perpendicular distances. Let $P(x, y, z)$ be the variable point.

- Distance d_1 from $x + y + z = 0$:

$$d_1 = \frac{|x + y + z|}{\sqrt{1^2 + 1^2 + 1^2}} = \frac{|x + y + z|}{\sqrt{3}} \implies d_1^2 = \frac{(x + y + z)^2}{3}$$

- Distance d_2 from $x - 2y + z = 0$:

$$d_2 = \frac{|x - 2y + z|}{\sqrt{1^2 + (-2)^2 + 1^2}} = \frac{|x - 2y + z|}{\sqrt{6}} \implies d_2^2 = \frac{(x - 2y + z)^2}{6}$$

- Distance d_3 from $x - z = 0$:

$$d_3 = \frac{|x - z|}{\sqrt{1^2 + 0^2 + (-1)^2}} = \frac{|x - z|}{\sqrt{2}} \implies d_3^2 = \frac{(x - z)^2}{2}$$

Step 2: Equate $d_1^2 + d_2^2 = d_3^2$.

$$\frac{(x + y + z)^2}{3} + \frac{(x - 2y + z)^2}{6} = \frac{(x - z)^2}{2}$$

Multiply the entire equation by 6:

$$2(x + y + z)^2 + (x - 2y + z)^2 = 3(x - z)^2$$

Step 3: Algebraic expansion and simplification. Let $u = x + z$. Then $x + y + z = u + y$ and $x - 2y + z = u - 2y$:

$$\begin{aligned} 2(u + y)^2 + (u - 2y)^2 &= 2(u^2 + 2uy + y^2) + (u^2 - 4uy + 4y^2) \\ &= 2u^2 + 4uy + 2y^2 + u^2 - 4uy + 4y^2 = 3u^2 + 6y^2 \end{aligned}$$

Now, rewrite $u^2 = (x + z)^2 = x^2 + 2xz + z^2$:

$$3u^2 + 6y^2 = 3(x^2 + 2xz + z^2) + 6y^2 = 3x^2 + 6xz + 3z^2 + 6y^2$$

The right-hand side is:

$$3(x - z)^2 = 3(x^2 - 2xz + z^2) = 3x^2 - 6xz + 3z^2$$

Equating both sides:

$$3x^2 + 6xz + 3z^2 + 6y^2 = 3x^2 - 6xz + 3z^2$$

Subtracting $3x^2 + 3z^2$ from both sides:

$$6xz + 6y^2 = -6xz \implies 6y^2 + 12xz = 0 \implies y^2 + 2xz = 0$$

Final Answer:

$$y^2 + 2xz = 0$$

28 Problem 28

Problem 28

Find the equations of the planes through the points $(0, 4, -3)$, $(6, -4, 3)$ other than the plane through the origin, which cut off from the axes intercepts whose sum is zero.

Step 1: Intercept form and given condition. Let the non-zero intercepts made by the plane on the coordinate axes be a, b, c .

$$\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$$

Given that the sum of the intercepts is zero:

$$a + b + c = 0 \implies c = -(a + b)$$

Thus the equation becomes:

$$\frac{x}{a} + \frac{y}{b} - \frac{z}{a+b} = 1 \quad \text{--- (1)}$$

Step 2: Substitute the two given points.

1. Point $(0, 4, -3)$:

$$\frac{0}{a} + \frac{4}{b} - \frac{-3}{a+b} = 1 \implies \frac{4}{b} + \frac{3}{a+b} = 1 \quad \text{--- (2)}$$

2. Point $(6, -4, 3)$:

$$\frac{6}{a} - \frac{4}{b} - \frac{3}{a+b} = 1 \implies \frac{6}{a} - \left(\frac{4}{b} + \frac{3}{a+b} \right) = 1 \quad \text{--- (3)}$$

From equation (2), substitute $\left(\frac{4}{b} + \frac{3}{a+b} \right) = 1$ into equation (3):

$$\frac{6}{a} - 1 = 1 \implies \frac{6}{a} = 2 \implies a = 3$$

Step 3: Solve for b and c . Substitute $a = 3$ into equation (2):

$$\frac{4}{b} + \frac{3}{3+b} = 1 \implies \frac{4(3+b) + 3b}{b(3+b)} = 1$$

$$12 + 4b + 3b = 3b + b^2 \implies b^2 - 4b - 12 = 0$$

Factoring:

$$(b-6)(b+2) = 0 \implies b = 6 \quad \text{or} \quad b = -2$$

Step 4: Find the equations for both cases.

- **Case 1:** $a = 3, b = 6$

$$c = -(a+b) = -(3+6) = -9.$$

$$\text{Equation: } \frac{x}{3} + \frac{y}{6} - \frac{z}{9} = 1.$$

Multiplying by 18:

$$6x + 3y - 2z = 18 \iff 6x + 3y - 2z - 18 = 0$$

- **Case 2:** $a = 3, b = -2$

$$c = -(a+b) = -(3-2) = -1.$$

$$\text{Equation: } \frac{x}{3} - \frac{y}{2} - \frac{z}{1} = 1.$$

Multiplying by 6:

$$2x - 3y - 6z = 6 \iff 2x - 3y - 6z - 6 = 0$$

Final Answer:

There are two such planes:

$$6x + 3y - 2z = 18$$

and

$$2x - 3y - 6z = 6$$

29 Problem 29

Problem 29

Find the equation of the plane through the line of intersection of the planes $ax + by + cz + d = 0$, $a_1x + b_1y + c_1z + d_1 = 0$ and parallel to the x -axis.

Step 1: Family of planes through the intersection. The family of planes passing through the intersection of the two planes is:

$$(ax + by + cz + d) + \lambda(a_1x + b_1y + c_1z + d_1) = 0$$

$$(a + \lambda a_1)x + (b + \lambda b_1)y + (c + \lambda c_1)z + (d + \lambda d_1) = 0$$

The normal vector of this plane is:

$$\mathbf{n} = (a + \lambda a_1)\hat{i} + (b + \lambda b_1)\hat{j} + (c + \lambda c_1)\hat{k}$$

Step 2: Condition for parallelism to the x -axis. The x -axis has direction vector $\hat{i} = (1, 0, 0)$. Since the plane is parallel to the x -axis, its normal vector $\mathbf{n}$ must be perpendicular to $\hat{i}$:

$$\mathbf{n} \cdot \hat{i} = 0 \implies a + \lambda a_1 = 0 \implies \lambda = -\frac{a}{a_1} \quad (a_1 \neq 0)$$

Step 3: Substitute λ and simplify.

$$(ax + by + cz + d) - \frac{a}{a_1}(a_1x + b_1y + c_1z + d_1) = 0$$

Multiplying by a_1 :

$$\begin{aligned} a_1(ax + by + cz + d) - a(a_1x + b_1y + c_1z + d_1) &= 0 \\ (aa_1 - aa_1)x + (a_1b - ab_1)y + (a_1c - ac_1)z + (a_1d - ad_1) &= 0 \\ (a_1b - ab_1)y + (a_1c - ac_1)z + (a_1d - ad_1) &= 0 \end{aligned}$$

Equivalently:

$$(ab_1 - a_1b)y + (ac_1 - a_1c)z + (ad_1 - a_1d) = 0$$

Final Answer:

$$(a_1b - ab_1)y + (a_1c - ac_1)z + (a_1d - ad_1) = 0$$

or equivalently: $(ab_1 - a_1b)y + (ac_1 - a_1c)z + (ad_1 - a_1d) = 0$

30 Problem 30

Problem 30

A triangle, the lengths of whose sides are a, b, c is placed so that the middle points of the sides are on the axes. Prove that the equation to its plane is

$$\frac{x}{\alpha} + \frac{y}{\beta} + \frac{z}{\gamma} = 1$$

Proof: Step 1: Coplanarity of the triangle and its midpoints.

Let the triangle be $\triangle PQR$ having side lengths:

$$QR = a, \quad RP = b, \quad PQ = c$$

Let D, E, F be the midpoints of the sides QR, RP, PQ respectively. Since the vertices P, Q, R lie in a plane Π , the midpoints D, E, F of its sides must also lie in the same plane Π .

Step 2: Coordinates of the midpoints on the axes.

The problem specifies that the midpoints of the sides lie on the coordinate axes:

- Midpoint D lies on the x -axis $\implies D(\alpha, 0, 0)$
- Midpoint E lies on the y -axis $\implies E(0, \beta, 0)$
- Midpoint F lies on the z -axis $\implies F(0, 0, \gamma)$

Since the three non-collinear points $D(\alpha, 0, 0), E(0, \beta, 0), F(0, 0, \gamma)$ lie on the coordinate axes and also lie in the plane Π , the plane Π has intercepts α, β, γ on the x, y, z axes respectively.

The equation of a plane in intercept form with intercepts α, β, γ is:

$$\frac{x}{\alpha} + \frac{y}{\beta} + \frac{z}{\gamma} = 1$$

Step 3: Relationship between intercepts α, β, γ and side lengths a, b, c .

By the Midpoint Theorem in plane geometry, the segment joining the midpoints of any two sides of a triangle is parallel to the third side and has length equal to half of that side:

$$EF = \frac{1}{2}QR = \frac{a}{2}, \quad FD = \frac{1}{2}RP = \frac{b}{2}, \quad DE = \frac{1}{2}PQ = \frac{c}{2}$$

Using the 3D distance formula between points $D(\alpha, 0, 0), E(0, \beta, 0), F(0, 0, \gamma)$:

$$EF^2 = (0 - 0)^2 + (\beta - 0)^2 + (0 - \gamma)^2 = \beta^2 + \gamma^2 = \frac{a^2}{4} \quad (4)$$

$$FD^2 = (\alpha - 0)^2 + (0 - 0)^2 + (0 - \gamma)^2 = \alpha^2 + \gamma^2 = \frac{b^2}{4} \quad (5)$$

$$DE^2 = (\alpha - 0)^2 + (0 - \beta)^2 + (0 - 0)^2 = \alpha^2 + \beta^2 = \frac{c^2}{4} \quad (6)$$

Summing equations (4), (5), and (6):

$$2(\alpha^2 + \beta^2 + \gamma^2) = \frac{a^2 + b^2 + c^2}{4} \implies \alpha^2 + \beta^2 + \gamma^2 = \frac{a^2 + b^2 + c^2}{8}$$

Subtracting equation (4) ($\beta^2 + \gamma^2 = \frac{a^2}{4}$):

$$\alpha^2 = \frac{a^2 + b^2 + c^2}{8} - \frac{a^2}{4} = \frac{b^2 + c^2 - a^2}{8} \iff 8\alpha^2 = b^2 + c^2 - a^2$$

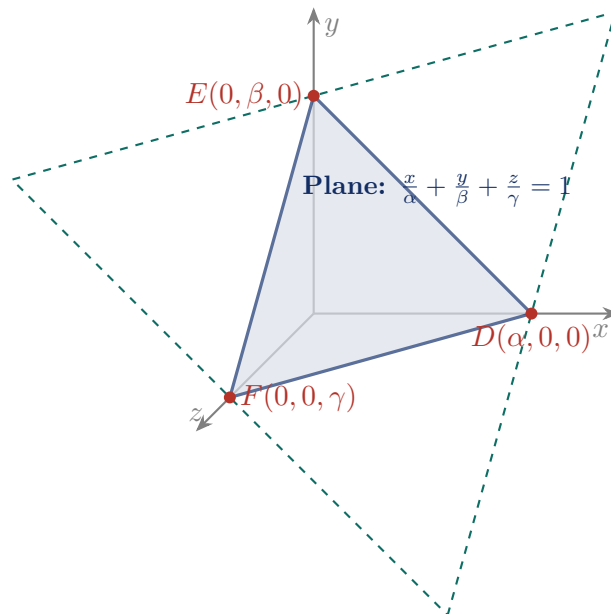
Similarly:

$$8\beta^2 = c^2 + a^2 - b^2, \quad 8\gamma^2 = a^2 + b^2 - c^2$$

This explicitly determines the intercepts α, β, γ in terms of the side lengths a, b, c . Hence, the equation to the plane of the triangle is:

$$\frac{x}{\alpha} + \frac{y}{\beta} + \frac{z}{\gamma} = 1$$

This completes the proof. □



Conclusion:

The midpoints of the sides lie in the plane of the triangle with intercepts α, β, γ on the coordinate axes. Therefore, the equation to its plane is:

$$\boxed{\frac{x}{\alpha} + \frac{y}{\beta} + \frac{z}{\gamma} = 1}$$

where $8\alpha^2 = b^2 + c^2 - a^2$, $8\beta^2 = c^2 + a^2 - b^2$, $8\gamma^2 = a^2 + b^2 - c^2$.